KRITI NYOUPANE

Project Manager · Systems Analyst · Software Developer

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PROFILE

IT professional with an MSc in Computer Engineering (Data Science and Analytics) and nearly three years at Venture23, first as a mid-level developer and then as project manager. Managed the development of an AI-driven teledermatology platform for an overseas client, working between the client and the development team on requirements, plans and progress. Before that, built retrieval-augmented generation (RAG) systems, knowledge graphs and an agentic AI application. Research background in API-based data pipelines, data modelling and data validation, with two first-author peer-reviewed papers, one accepted for IEEE Computer Society proceedings.

CORE SKILLS

Project delivery	Project planning and scheduling; scope, milestones and estimates; risk and dependency management; resource planning and work allocation; stakeholder coordination; progress and status reporting; user acceptance, deployment and handover
Systems analysis	Requirements elicitation and stakeholder interviews; requirements specification (SRS); process modelling (UML, data flow diagrams); data modelling (ER models, knowledge graphs); API analysis and integration; acceptance criteria and test scenarios; technical documentation
Technical	Python, C, C++; LLM applications (RAG, knowledge graphs, agentic AI); machine learning (Random Forest, XGBoost, LightGBM); deep learning (CNN, U-Net, BERT, NeRF); geospatial data (GeoTIFF rasters); GPU-accelerated training; front-end development; Git and GitHub

PROFESSIONAL EXPERIENCE

Venture23, Kathmandu, Nepal 26 June 2023 – 30 April 2026

Project Manager

- Managed the development of an AI-driven teledermatology platform for SkinHappy, an overseas client.
- Served as the link between the client and the development team: gathered and clarified requirements, agreed priorities and acceptance criteria, and recorded decisions and changes in scope.
- Planned milestones, allocated work across the team, and tracked development, testing and defect resolution through regular reviews.
- Reported progress, risks and open issues to the client and to management, and coordinated releases with the client.

Mid-Level Developer (AI/ML)

- Built retrieval-augmented generation (RAG) systems and designed knowledge graphs that model the entities and relationships the applications rely on.
- Developed myOS, an agentic AI system for personal development.

RESEARCH AND TECHNICAL EXPERIENCE

MSc Thesis Research, IOE Pulchowk Campus, Tribhuvan University 2024 – 2026

Multi-output machine learning for site-specific fertilizer recommendation in Nepal

- Built an automated data pipeline on the NARC/NSSRC Soil Information API and GeoTIFF raster layers, producing a nationwide soil-to-fertilizer dataset of 384,077 records covering all 77 districts.
- Trained multi-output models for 40 fertilizer targets (Random Forest $R^2 = 0.983$) and benchmarked CPU against GPU training (XGBoost 7.4× faster on GPU).
- Validated the API's historic soil data against 2026 field measurements and documented where it had drifted and needed fresh local data.

Indian Institute of Technology (IIT) Roorkee, India April 2022 – May 2022

Research Intern, SPARK International Research Internship Programme

- Worked with a PhD scholar on Android malware classification using deep image learning (DIDroid): processed benign and malicious APKs across 14 malware categories, extracted code features with Apktool, and ran ablations on a 1D CNN.

Impacters (Project Saathi), Nepal April 2021 – September 2021

Research Machine Learning Engineer

- Built a crop recommendation system for Nepali farms that combined sensor readings (temperature, humidity, moisture) with soil types classified from images.
- Defined the soil classification scheme with agricultural experts and merged sensor and image features into a single dataset.
- Benchmarked CNN, MobileNet and U-Net against classical machine learning models, and co-authored an applied article on the work.

Academic Projects, BE Computer Engineering 2018 – 2023

- Major project: 3D reconstruction for a virtual tour using Neural Radiance Fields (NeRF). Minor project: transformer-based text similarity search using BERT Sentence Transformers.

EDUCATION

MSc in Computer Engineering (Data Science and Analytics) May 2024 – May 2026

Institute of Engineering, Pulchowk Campus, Tribhuvan University, Lalitpur

Aggregate 86.53%. Studied on an IOE merit scholarship.

Bachelor of Engineering in Computer Engineering November 2018 – April 2023

Institute of Engineering, Pulchowk Campus, Tribhuvan University, Lalitpur

Aggregate 72.3%. Ranked in the top 500 of about 16,000 candidates in the 2018 IOE entrance examination.

Higher Secondary Education (+2) 2015 – 2017

The Times International College, Dillibazar, Kathmandu

81.3%. Mahatma Gandhi Scholarship, Embassy of India, Kathmandu.

PUBLICATIONS

1. Nyoupane, K., Giri, S., Ito, A., Parajuli, R., Purbey, J., Thapa, S. & Chaulagain, B. (2026). A multi-output machine learning framework for site-specific fertilizer quantity recommendation reflecting Nepal's diverse agricultural landscape. *CANDARW 2026, 11th International Workshop on GPU Computing and AI (GCA)*, IEEE Computer Society. In press; to be presented in Ishigaki, Japan, 24–27 November 2026.
2. Nyoupane, K., Giri, S. & Purbey, J. (2026). Predicting fertilizer quantity using geo-spatial and physicochemical properties of soil. *8th KEC Conference on Engineering and Technology*, Lalitpur, Nepal. Presented 7 May 2026.
3. Pokharel, N. & Nyoupane, K. (2021). Discerning soil properties using a convolutional neural network. *Impacters* (applied article).

TRAINING AND AWARDS

2024 Merit scholarship for MSc, IOE, Tribhuvan University

2022 AI & Deep Learning Fellowship (Microdegree), Fusemachines

2022 SPARK International Research Internship Scholarship, IIT Roorkee

2022 Third Runner-Up, Datarush ML competition, LOCUS

2021 Scholar, Annual Nepal AI School (ANAIS), NAAMII

LEADERSHIP AND VOLUNTEERING

President, IOE Pulchowk Girls (women in STEM)

Founding Member, IT Club, Pulchowk Campus

Community Builder, NxtGen (startup), 2020–2021

Volunteer, Team Nepo (youth empowerment and nature conservation)

LANGUAGES

Nepali (native); English (professional working proficiency); Hindi (proficient)

REFERENCES

Available on request.